



Refining the Sunspot Number Series to ISN V3: Challenges and Benefits for the Space Climate Community

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Abstract

The International Sunspot Number (ISN) series, a cornerstone for space climate studies, still exhibits scale discrepancies despite recent recalibrations, underscoring the critical need for a comprehensive reconstruction from raw historical observations. The FARSuN (Findability and Accessibility of historical Raw Sunspot Numbers) project addresses this by systematically gathering, digitizing, and standardizing scattered sunspot data from 1607-1980, aiming to bridge the technical and scientific gap preventing holistic data use. Significant progress includes the digitization of Prof. Wolf's Mittheilungen (1610-1918) and the near-completion (95%) of more than 2000 Zürich observation tables, transforming them into machine-readable formats. Data extractions for challenging sources like Gruithuisen, Stark, Adams, and Rost are actively underway, supported by advanced tools like Transkribus and rigorous quality control efforts to define homogeneous metadata criteria. Upcoming initiatives include a citizen science web interface to engage the public in data recovery and the ultimate goal of constructing a new, consolidated ISN (V3) and historical hemispheric sunspot numbers. This comprehensive reconstruction will provide the Space Climate Community with an unprecedentedly accurate and accessible record, significantly improving solar cycle predictions, understanding long-term solar variability, space weather discussions, and refining climate models.

Why Reconstruction?

The International Sunspot Number (ISN) suffers from historical inconsistencies, like the "Waldmeier jump" around 1947. Previous attempts to fix this were limited to *recalibration*. FARSuN's approach, a full **reconstruction**, is fundamentally different and more robust.

- Recalibration adjusts an existing index, biases may persist.
- Reconstruction returns to primary sources (Historical journals, drawings, manuscripts, tables).
- Tackles observer-dependent scale changes ("Waldmeier jump" from ?)

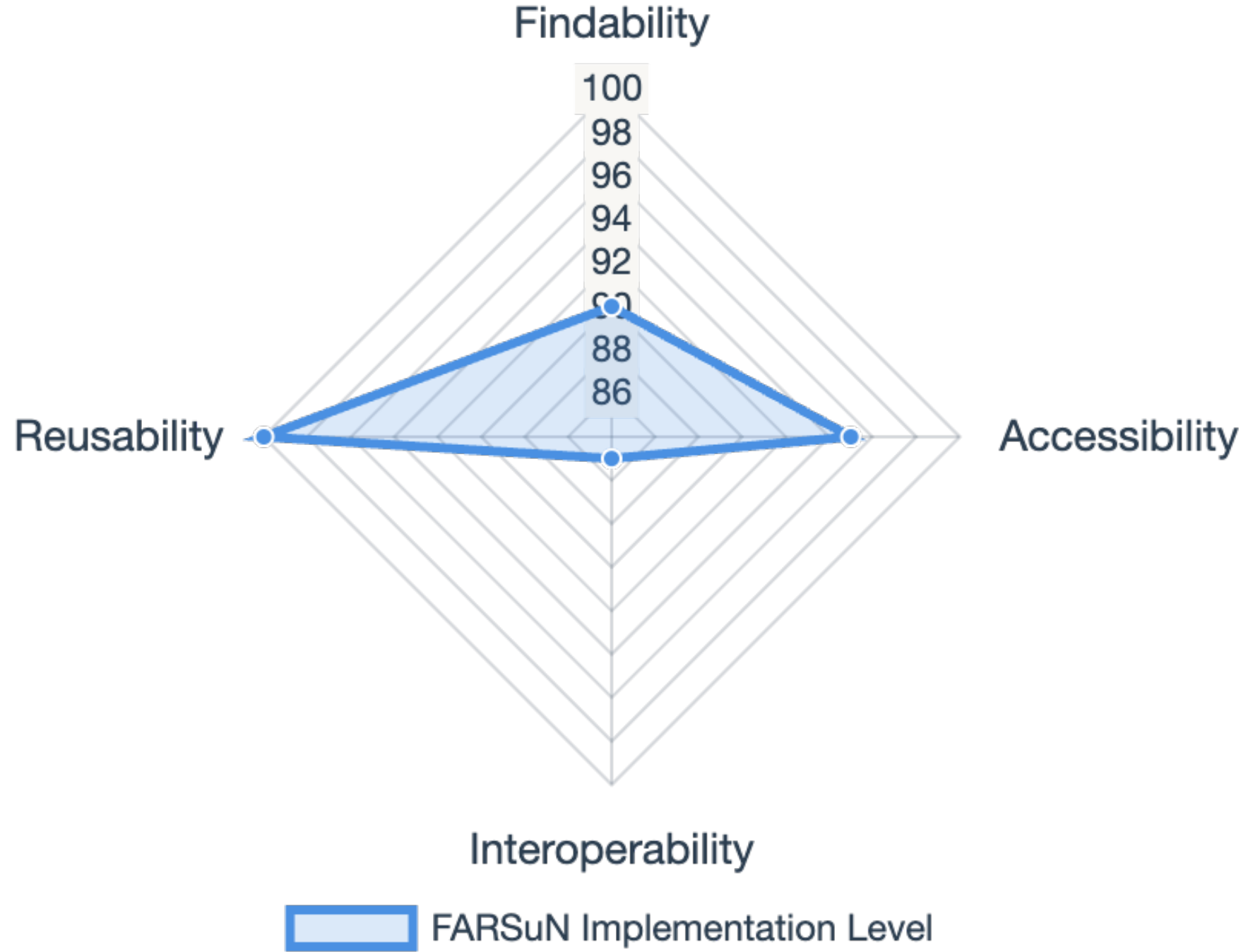
Four-Axis Methodology

1. **Gather Sources** : archives, manuscripts, tables (Identify and locate historical sunspot data)
2. **Process Data** : digitize, OCR/HWR, Transkribus (Extract and normalize data)
3. **Validate & Standardize** : homogeneous metadata for scientific usability
4. **Disseminate** : FAIR principles, VO-ready access

FAIR Principles

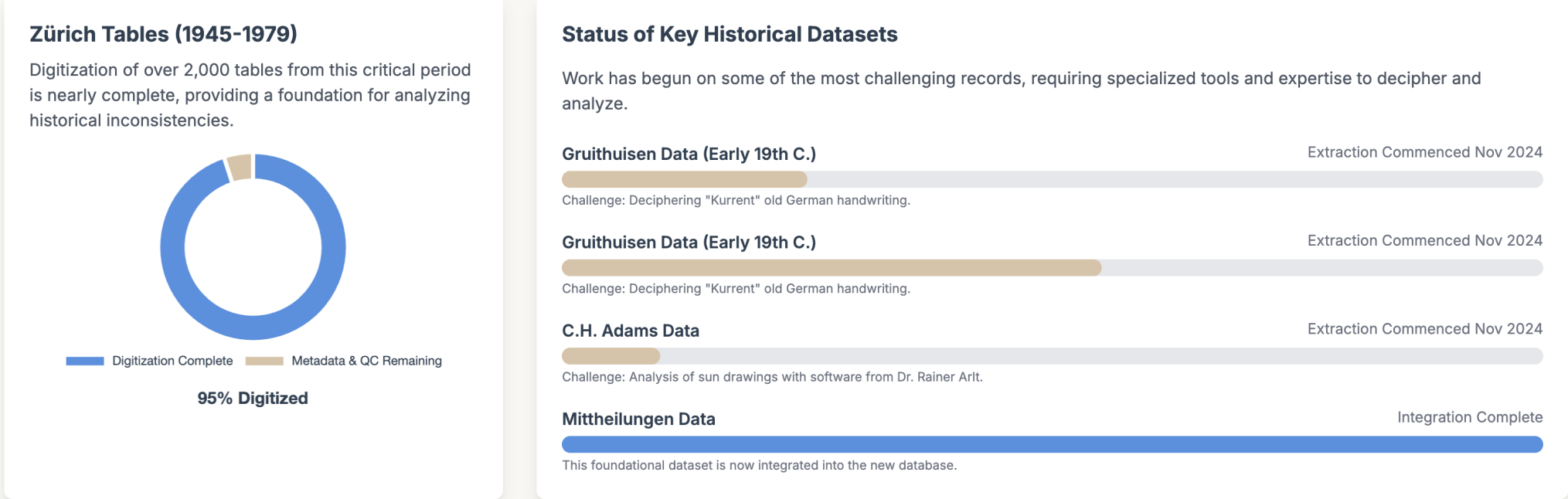
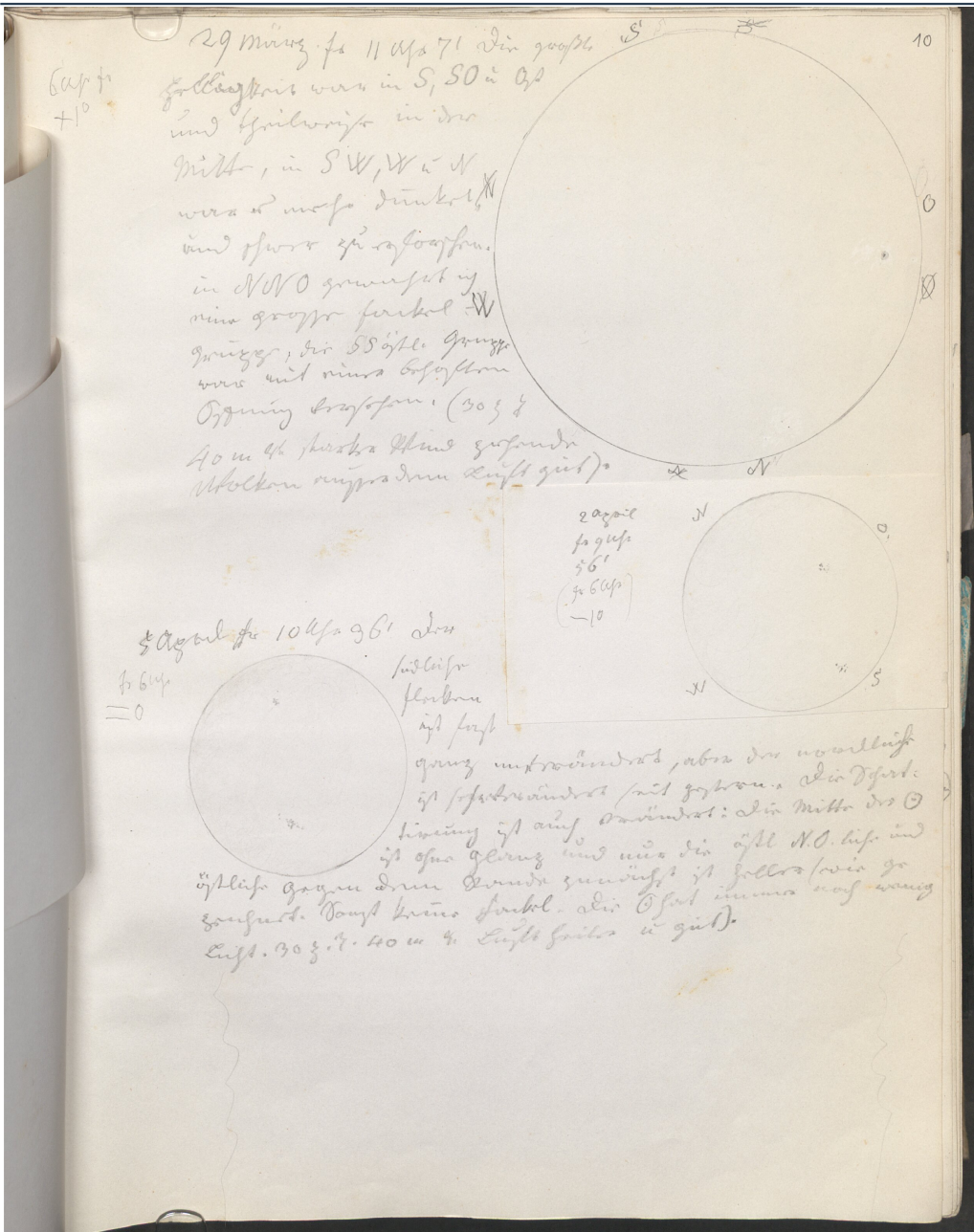
The project ensures FAIR (Findable, Accessible, Interoperable, Reusable) principles, making the final dataset robust and accessible for future science.

- **Findable** – DOIs, metadata
- **Accessible** – open formats
- **Interoperable** – vocabularies, VO compliance
- **Reusable** – versioning, uncertainties

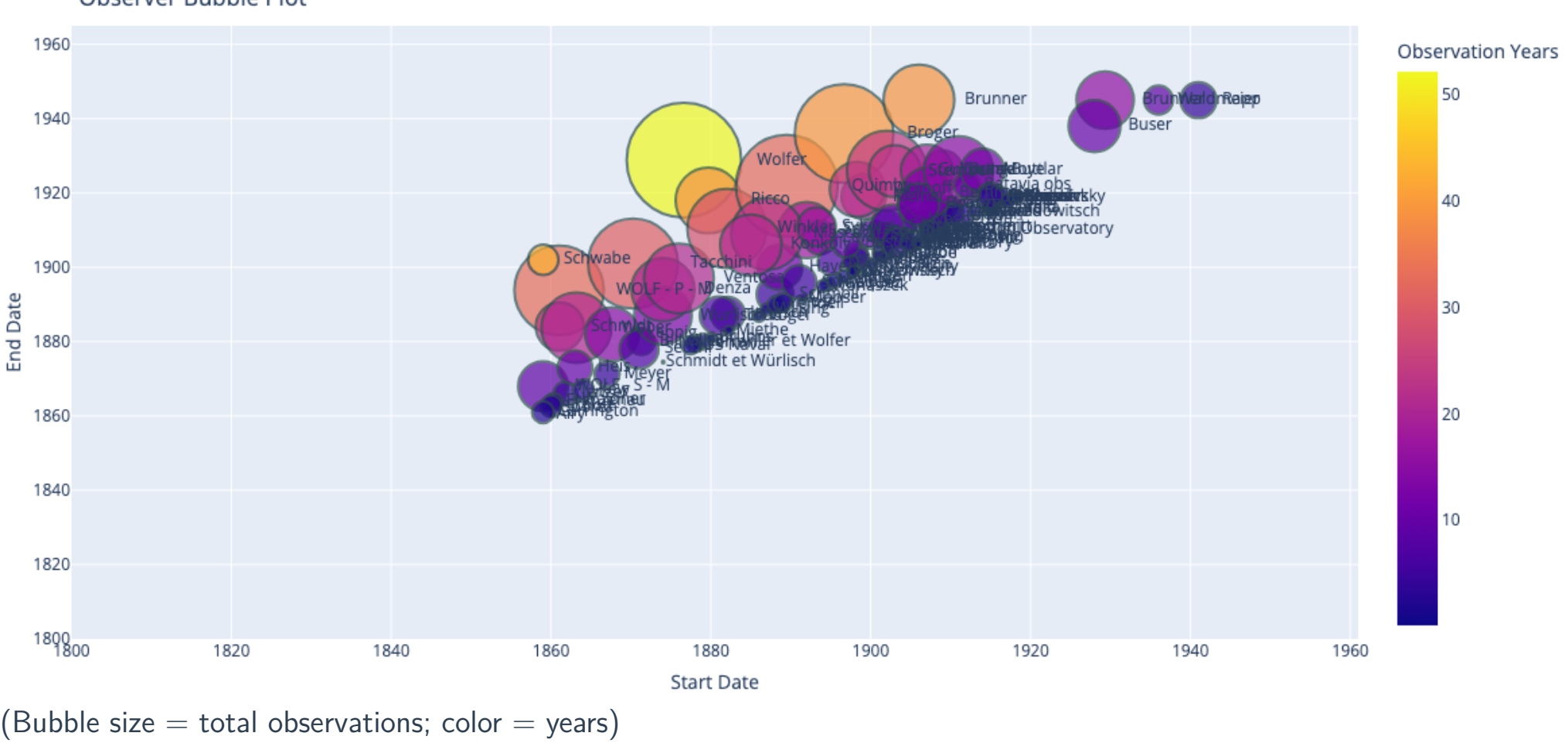


Key Datasets

Source	Period	Status
Zürich Tables	1945–1979	95% digitized
Mittheilungen	1610–1918	Integrated indices
Gruithuisen	1800s	In progress (<i>Kurrent</i>)
Adams	19–20c	Drawings under extraction
Stark	18–19c	Tabular records
Rost	18–19c	Metadata normalization



Observer Timeline



Methods

- Transcription: OCR/HWR + Transkribus
- Normalization: unified dates, observer IDs
- Uncertainties: propagate errors
- Reproducibility: scripted pipelines

From Raw to ISN V3

1. Curate daily counts
2. Calibrate observers
3. Aggregate to monthly/yearly
4. Compare vs. SN V2

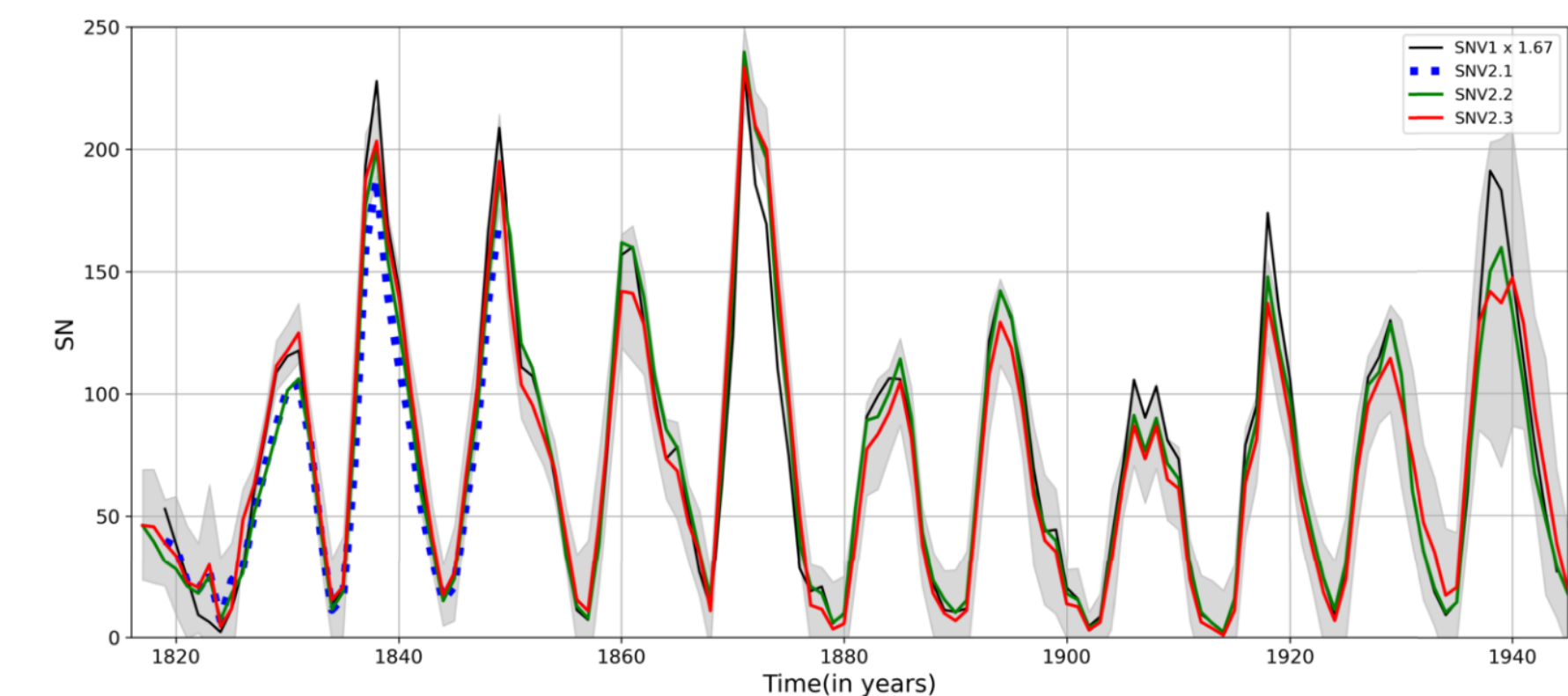


Figure 15 Annually averaged SN series. The grey-shaded area represents the 1σ uncertainty of SNV2.3. We note that SNV1 has been multiplied by 1.67 to bring it to the same scale as the other series.

Challenges

- Handwriting (*Kurrent*)
- Heterogeneous formats
- Observer bias
- Provenance tracking

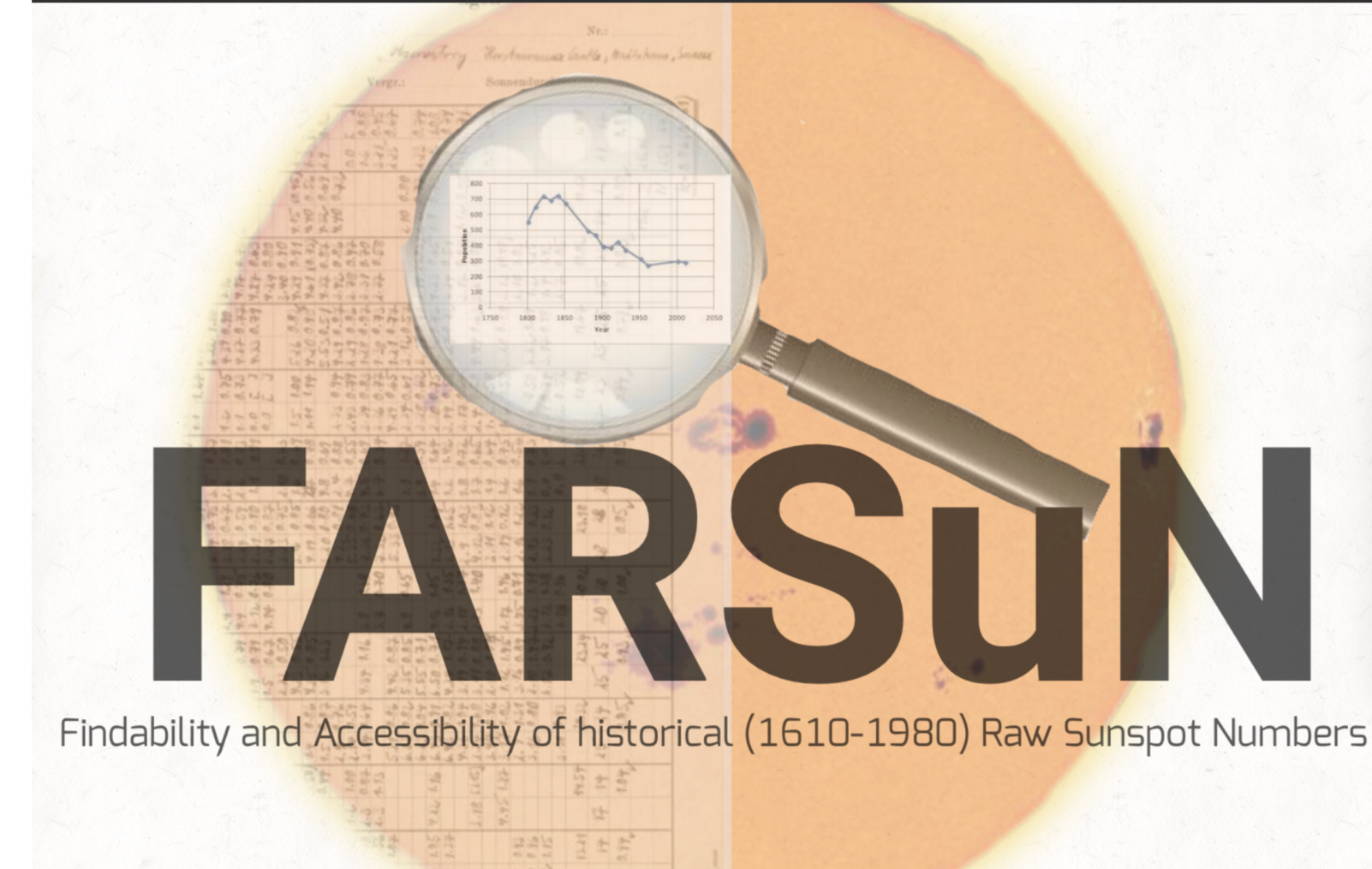
Impact

- Better forecasts
- Long-term variability
- Space weather baselines
- Climate model inputs

Outlook

- Citizen science portal
- Data releases + VO access
- Collaboration open

Findability and Accessibility of historical (1610-1980) Raw Sunspot Numbers



References